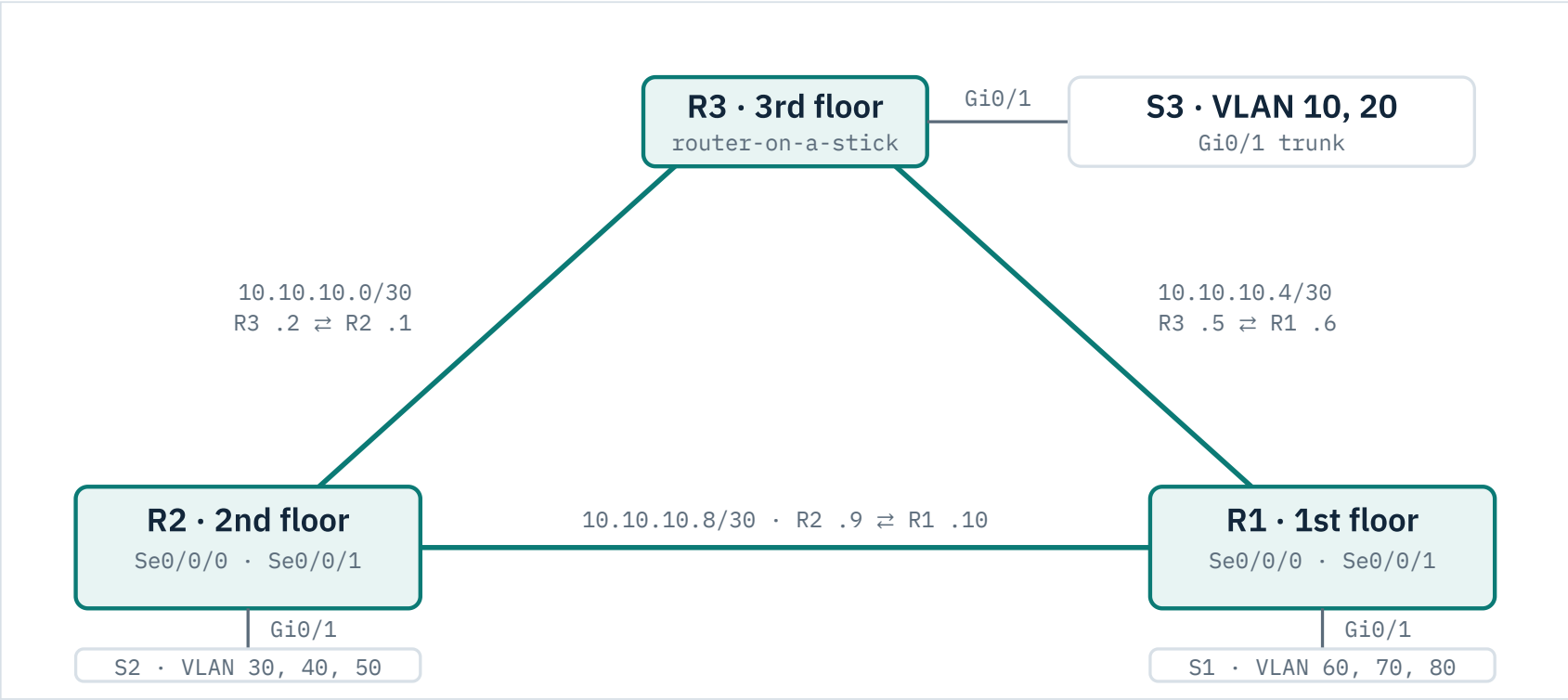


# Hotel Network Documentation

Three-floor campus network: one router and one switch per floor, joined by a routed OSPF triangle.

|                                  |                                  |                                  |                                       |
|----------------------------------|----------------------------------|----------------------------------|---------------------------------------|
| <b>3</b><br>Routers (Cisco 2911) | <b>3</b><br>Switches (2960-24TT) | <b>8</b><br>VLANs / user subnets | <b>OSPF area 0</b><br>Dynamic routing |
|----------------------------------|----------------------------------|----------------------------------|---------------------------------------|

## Topology



## Router-to-router links (point-to-point /30)

| Link    | Subnet     | Mask            | End A           | End B            | Clock (DCE) |
|---------|------------|-----------------|-----------------|------------------|-------------|
| R3 ↔ R2 | 10.10.10.0 | 255.255.255.252 | R2 Se0/0/0 · .1 | R3 Se0/0/0 · .2  | R2          |
| R3 ↔ R1 | 10.10.10.4 | 255.255.255.252 | R3 Se0/0/1 · .5 | R1 Se0/0/0 · .6  | R3          |
| R2 ↔ R1 | 10.10.10.8 | 255.255.255.252 | R2 Se0/0/1 · .9 | R1 Se0/0/1 · .10 | R1          |

## VLANs and subnets

| Floor         | VLAN | Name      | Subnet         | Mask          | Gateway (sub-interface) | Switch ports          |
|---------------|------|-----------|----------------|---------------|-------------------------|-----------------------|
| 3rd · R3 / S3 | 10   | IT        | 192.168.1.0/24 | 255.255.255.0 | 192.168.1.1 · Gi0/1.10  | S3 Fa0/1-2, 0/6, 0/20 |
|               | 20   | Admin     | 192.168.2.0/24 | 255.255.255.0 | 192.168.2.1 · Gi0/1.20  | S3 Fa0/3-5            |
| 2nd · R2 / S2 | 30   | Sales     | 192.168.3.0/24 | 255.255.255.0 | 192.168.3.1 · Gi0/1.30  | S2 Fa0/3-5            |
|               | 40   | HR        | 192.168.4.0/24 | 255.255.255.0 | 192.168.4.1 · Gi0/1.40  | S2 Fa0/1-2            |
|               | 50   | Finance   | 192.168.5.0/24 | 255.255.255.0 | 192.168.5.1 · Gi0/1.50  | S2 Fa0/6-7            |
| 1st · R1 / S1 | 60   | Logistics | 192.168.6.0/24 | 255.255.255.0 | 192.168.6.1 · Gi0/1.60  | S1 Fa0/1-3            |
|               | 70   | Store     | 192.168.7.0/24 | 255.255.255.0 | 192.168.7.1 · Gi0/1.70  | S1 Fa0/4-5            |
|               | 80   | Reception | 192.168.8.0/24 | 255.255.255.0 | 192.168.8.1 · Gi0/1.80  | S1 Fa0/6-7            |

Each switch's Gi0/1 is an 802.1Q trunk to its router's Gi0/1. Unused ports stay in VLAN 1.

Services

| Service       | Detail                                                                                                            |
|---------------|-------------------------------------------------------------------------------------------------------------------|
| DHCP          | Each router serves its own VLANs (pools POOL-10 to POOL-80). The .1 gateway is excluded from every pool.          |
| Hotel server  | 192.168.1.150 (VLAN 10, excluded from DHCP). Used as the NTP server on all devices except R2.                     |
| Routing       | OSPF area 0 on all three routers: R3 (router-id 3.3.3.3), R2 (2.2.2.2), R1 (1.1.1.1). Process IDs are 1, 1 and 2. |
| Remote access | SSH only on router VTY lines with local users (domain network.local).                                             |
| Wi-Fi         | Access points on each floor (Admin, Sales, Logistics) carry laptops, tablets and phones on that floor's VLAN.     |
| Port security | S3 Fa0/20: sticky MAC 0001.9697.7706.                                                                             |

Access control

| Router | ACL 100                                                                                        | Applied inbound on              |
|--------|------------------------------------------------------------------------------------------------|---------------------------------|
| R2     | deny icmp 192.168.3.0/24 → host 192.168.1.150<br>deny icmp 192.168.4.0/24 → host 192.168.1.150 | Gi0/1.30 (Sales), Gi0/1.40 (HR) |
| R1     | permit tcp 192.168.7.0/24 → host 192.168.1.150 eq 443                                          | Gi0/1, Gi0/1.70 (Store)         |